# DEVELOPMENT AND ASSESSMENT OF A SMART NSTP MANAGEMENT AND AI-INTEGRATED LEARNING PLATFORM FOR TARLAC AGRICULTURAL UNIVERSITY–NSTP OFFICE

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# Chapter 1

## THE PROBLEM AND ITS BACKGROUND

This chapter presents the problem and background of the study on the development and assessment of a Smart NSTP Management and AI-Integrated Learning Platform for the Tarlac Agricultural University-National Service Training Program (TAU-NSTP) Office. It discusses the rationale, reviews and synthesizes relevant literature, explains the conceptual framework, states the research problem, establishes the scope and delimitation, identifies the significance of the study, and defines the principal terms used in the investigation.

### Rationale 1

The National Service Training Program (NSTP) is a major co-curricular responsibility of Philippine higher education institutions. At Tarlac Agricultural University (TAU), its Civic Welfare Training Service (CWTS), Literacy Training Service (LTS), and Reserve Officers' Training Corps (ROTC) components connect classroom preparation with civic engagement, literacy, community development, discipline, and public service. The scale of these activities creates a continuing need to manage student registration, component and section assignments, attendance, facilitator assignments, learning materials, assessments, grades, announcements, supporting documents, and regulatory reports accurately and on time. The University's 2026 CWTS and LTS graduation, involving 1,104 completers, illustrates the volume of records and coordination that the NSTP Office may handle within a single academic cycle (Tarlac Agricultural University, 2026).

The administrative environment of NSTP is also becoming more data intensive. In Central Luzon, the Commission on Higher Education Regional Office III directed higher education institutions to submit NSTP graduate data electronically and in prescribed spreadsheet formats to facilitate the processing of serial numbers (Commission on Higher Education Regional Office III [CHEDRO III], 2022a, 2022b). More recent regional memoranda have continued to require timely NSTP implementation and monitoring data, including institution-level surveys and reports (CHEDRO III, 2024, 2025). These requirements show that dependable local records are not only operational conveniences; they are inputs to institutional accountability and external reporting. A fragmented workflow makes it difficult to maintain one verified version of student, attendance, grade, and completion data across multiple components and sections.

Based on the preliminary problem context presented by the proponent, several TAU-NSTP processes remain manual or distributed across separate files, forms, communication channels, and individual work practices. Registration and sectioning may require repeated encoding; attendance sheets and grades may need to be consolidated; materials and announcements may be sent through different channels; and reports may have to be reconstructed from records held by different personnel. Such fragmentation can increase processing time, produce duplicate or inconsistent entries, delay retrieval and reporting, and make the status of a student or section difficult to see at a glance. It can also place unnecessary clerical demands on administrators, coordinators, and facilitators whose primary responsibilities include program delivery and student development.

An integrated platform offers a practical response because administrative and learning activities can share a controlled database and a common set of user roles. Evidence

on successful e-learning systems indicates that system quality, information quality, service and support, the educational experience, and the characteristics of learners and instructors jointly affect satisfaction, use, and perceived benefits (Al-Fraihat et al., 2020). A platform for NSTP should therefore do more than digitize paper forms. It should provide accurate information, support the actual work of each user group, reduce unnecessary steps, and make learning and administrative services accessible through a consistent interface.

Attendance is an appropriate area for targeted automation because it is frequent, time sensitive, and directly connected to student status and reporting. QR-based attendance systems can reduce the effort required to record and transmit attendance compared with manual sign-in procedures (Rahmawati et al., 2020). Within the proposed platform, a student-specific QR identifier and controlled scanning workflow can link attendance events to the correct student, component, section, facilitator, date, and time. However, convenience must be paired with safeguards such as authenticated user sessions, time-bounded attendance windows, duplicate-scan prevention, audit logs, and facilitator verification so that the resulting records are credible and correctable.

The learning component is equally important. Centralized lessons, supplementary readings, activities, quizzes, submissions, scores, and progress records can give students a clearer learning pathway while allowing facilitators to monitor participation and performance. Learning analytics research shows that data from learning management systems, assessments, and student interactions can support feedback, dashboards, and carefully designed interventions, although evidence must be interpreted cautiously and the educational purpose must remain explicit (Ifenthaler & Yau, 2020; Mangaroska & Giannakos, 2019). The proposed system therefore treats progress data as support for

facilitator judgment and student guidance rather than as an automatic determination of ability or final standing.

Artificial intelligence (AI) adds the possibility of immediate, context-aware assistance. Systematic reviews report that educational chatbots have been used for teaching, administrative assistance, assessment, and advising, and may improve access to information and provide individualized support (Kuhail et al., 2023; Labadze et al., 2023; Okonkwo & Ade-Ibijola, 2021). For TAU-NSTP, a chatbot grounded in approved program information and learning materials could answer routine questions, explain requirements, direct students to resources, and assist users outside office hours. A project proposal advisor could guide students through problem identification, objectives, beneficiaries, activities, resources, timelines, risks, and expected outcomes without replacing student authorship or facilitator approval.

AI may also help personalize recommended learning resources and generate preliminary rubric-based feedback. Reviews of AI-supported student assessment and generative-AI feedback describe potential benefits in timely feedback and workload support, but they also identify concerns about validity, bias, accuracy, transparency, and overreliance (Gonzalez-Calatayud et al., 2021; Lee & Moore, 2024; Yan et al., 2024). Consequently, the proposed platform will use AI as an assistive layer: recommendations and draft feedback must be traceable to approved content or predefined rubrics, communicated as guidance, and subject to human review. Facilitators will retain authority over official evaluations and grades.

The concentration of student profiles, educational records, attendance histories, submissions, and AI interaction data creates an obligation to protect privacy and security.

The Data Privacy Act of 2012 requires personal information processing to observe lawful purpose, transparency, proportionality, and appropriate protection (Republic of the Philippines, 2012). UNESCO's guidance on generative AI likewise calls for a human-centered approach that protects agency, inclusion, privacy, and educational integrity (UNESCO, 2023). These principles require role-based access, data minimization, informed notices, secure authentication, access and change logs, retention controls, backup and recovery procedures, and clear rules on what data may be sent to an AI service.

System development must finally be accompanied by systematic assessment. ISO/IEC 25010:2023 provides a current product-quality model composed of nine characteristics: functional suitability, performance efficiency, compatibility, interaction capability, reliability, security, maintainability, flexibility, and safety (International Organization for Standardization & International Electrotechnical Commission [ISO/IEC], 2023). Applying relevant characteristics during requirements definition, testing, expert review, and user evaluation helps connect the platform's features with measurable quality expectations. User perceptions must also be examined because performance expectancy, effort expectancy, facilitating conditions, and related acceptance factors influence whether a technology is adopted in practice (Venkatesh et al., 2012).

In view of the administrative demands, learning-support opportunities, regulatory expectations, and risks described above, the study proposes the development and assessment of a Smart NSTP Management and AI-Integrated Learning Platform for the TAU-NSTP Office. The intended contribution is a single, role-based environment that supports registration and sectioning, attendance, records, learning resources, assessment, grade computation, communication, and reporting while incorporating bounded AI

assistance. The study will document actual stakeholder requirements, iteratively develop the platform, evaluate its quality and acceptability, and identify improvements needed for responsible institutional use.

## Literature Review 6

### *Digital Administration and NSTP Reporting*

Digital transformation in education is not limited to online instruction; it also changes how institutions collect, validate, exchange, and report data. For NSTP offices, administrative information is closely tied to the delivery of CWTS, LTS, and ROTC. Registration establishes who belongs to a component and section; attendance documents participation; assessment and grade records show completion; and consolidated data support institutional and regulatory reports. CHEDRO III's shift to electronic NSTP submissions demonstrates that local records must be available in structured, timely, and reusable forms (CHEDRO III, 2022a, 2022b). The 2024 and 2025 NSTP data-collection directives further show that monitoring is recurring rather than a one-time requirement (CHEDRO III, 2024, 2025).

A centralized information system can reduce the distance between an operational event and an official report. When enrollment, section assignment, attendance, grades, and completion status are connected through common identifiers, authorized personnel can retrieve a student's history and produce summaries without repeatedly reconciling separate files. This arrangement also supports data validation because duplicate entries, missing fields, and inconsistent status values can be detected at the point of entry. The literature on e-learning system success reinforces that information quality and system quality are

foundational: users are more likely to benefit when information is accurate, relevant, understandable, and available through a dependable system (Al-Fraihat et al., 2020).

The TAU context makes scale and coordination particularly relevant. The University reported 1,104 CWTS and LTS completers in May 2026, apart from the administrative needs of ROTC and students in earlier stages of the program (Tarlac Agricultural University, 2026). This volume implies numerous recurring transactions across students, facilitators, sections, and activities. The literature and policy context therefore support an integrated platform, but they also indicate that digitization must be based on verified office workflows. Automating an unclear or redundant process can preserve the same weakness in digital form; needs assessment and stakeholder validation must precede feature implementation.

### *Integrated Student and Learning Management*

The proposed platform combines features normally associated with student information systems and learning management systems. The administrative layer maintains profiles, sections, attendance, grades, and reports, while the learning layer distributes content, receives submissions, administers assessments, and records progress. Combining these layers can reduce duplicate accounts and disconnected records, but integration should be purposeful. Each item of data should have a defined owner, source, validation rule, access level, and use. For example, a facilitator may record or verify section attendance, while only authorized personnel may change a student's official component assignment.

Al-Fraihat et al. (2020) found that successful e-learning systems depend on a broad set of interacting quality conditions rather than a single feature. Their model highlights system, information, service, educational, support, learner, and instructor factors. This

finding is important to the present study because an NSTP platform will be used by people with different responsibilities and levels of technical confidence. A function may be technically correct but still fail if its instructions are unclear, its outputs do not match office forms, it creates additional steps, or users cannot obtain assistance. Requirements must therefore cover workflow fit, clarity, accessibility, support, and recoverability as well as functional completeness.

An integrated learning environment can also make program content more consistently available. Lessons, readings, activity instructions, rubrics, due dates, announcements, and feedback can be organized by component and section. Students gain one place to view requirements and progress, while facilitators gain a shared structure for distributing materials and monitoring submissions. The platform should nevertheless allow authorized variation across CWTS, LTS, and ROTC, because the components have distinct activities and administrative arrangements. Integration in this sense means a common architecture and data model, not the elimination of legitimate program differences.

### *QR-Based Attendance and Record Integrity*

QR technology is widely used as a low-cost link between a physical identifier or displayed code and a digital record. Rahmawati et al. (2020) developed a mobile QR attendance system in which time information was encoded and scans were transmitted to a server, showing how a manual classroom procedure can be converted into a rapid digital transaction. For NSTP, QR-based attendance is suitable because students may participate in scheduled sessions, field activities, orientations, and component-specific events that require an auditable record.

A QR code alone, however, does not guarantee authentic attendance. A static image may be copied or shared, a scan may occur outside the intended time or venue, and an incorrect assignment may link the event to the wrong section. The proposed design should therefore treat the QR code as one element in a controlled workflow. Student authentication, server-side validation, event and section matching, scan timestamps, configurable attendance windows, duplicate prevention, audit trails, and facilitator review can strengthen record integrity. The system should also provide a documented correction process because legitimate exceptions, connectivity problems, and encoding errors may occur.

The most useful attendance system is connected to other NSTP records. Verified scans can update attendance histories, present-late-absent summaries, facilitator dashboards, and printable reports without re-encoding. This linkage reduces clerical effort and makes discrepancies visible sooner. At the same time, attendance data should not be used for unrelated monitoring, and location or device data should not be collected unless a documented need, lawful basis, and proportionate safeguard exist.

### *Learning Analytics and Personalized Support*

Digital learning systems produce traces such as assessment results, submission status, resource access, and task completion. Learning analytics organizes such data to support understanding and improvement of learning. Ifenthaler and Yau (2020) found that analytics may support dashboards, alerts, feedback, and adaptive interventions, but they also noted limited rigorous evidence and the importance of theory, institutional capability, ethics, and privacy. Mangaroska and Giannakos (2019) similarly emphasized the

connection between learning analytics and learning design: data become educationally useful when they inform purposeful decisions about activities, support, and feedback.

For the proposed platform, personalization will be bounded by the NSTP curriculum and by facilitator-approved resources. Assessment results and progress indicators may be used to recommend a review lesson, supplementary reading, practice activity, or consultation. The recommendation should explain the observed basis, such as a low score in a topic or an incomplete prerequisite, rather than presenting an unexplained prediction. Students and facilitators should be able to view, correct, or contest underlying data when appropriate. The platform will not label students permanently or make high-stakes decisions solely from an algorithmic score.

This approach addresses a central tension in personalized learning. Automation can help deliver timely and differentiated support to many students, but personalization based on poor data or opaque rules can misdirect learners. The literature supports the use of analytics as a decision-support mechanism, with educators retaining responsibility for interpreting context and choosing interventions (Ifenthaler & Yau, 2020). Accordingly, the study will evaluate whether recommendations are relevant, understandable, and useful to intended users, not merely whether the system can generate them.

### *AI Chatbots and Project Proposal Guidance*

Educational chatbots provide conversational access to information and learning support. Okonkwo and Ade-Ibijola's (2021) systematic review found chatbot applications in teaching and learning, administration, assessment, and advising. Kuhail et al. (2023) reported that educational chatbots have been implemented in different interaction styles and roles, often as teaching or peer agents, with evidence commonly focused on satisfaction

and learning outcomes. Labadze et al. (2023) further identified benefits for students in homework and study support, personalized learning, and skills development, while educators may benefit from time savings and pedagogical assistance.

These findings support a chatbot for routine NSTP inquiries, but the reported limitations are equally important. A general-purpose model may produce fluent but inaccurate answers, fail to reflect current local policies, or respond beyond its authorized scope. The proposed chatbot should therefore be grounded in a curated knowledge base containing approved office information, frequently asked questions, schedules, program guidelines, and learning materials. Responses should identify their informational nature, provide links or references to authoritative content when available, and direct high-risk, personal, or unresolved concerns to an authorized staff member.

Project proposal guidance is a related but distinct function. NSTP students may need help translating a community concern into a coherent proposal with objectives, beneficiaries, activities, resources, timelines, risk controls, and evaluation indicators. AI can use structured prompts and a facilitator-approved template to ask clarifying questions, identify missing sections, and suggest revisions. It should not invent community data, guarantee feasibility, or produce an official proposal without student verification. The student remains the author, and the facilitator retains responsibility for academic and program approval.

### *AI-Assisted Assessment and Feedback*

AI-supported assessment includes automated scoring, classification, feedback generation, and analysis of student responses. Gonzalez-Calatayud et al. (2021) found that AI applications in student assessment were concentrated in tutoring and assessment

contexts and called for educator preparation and further evidence. More recent work on generative AI has expanded interest in immediate, individualized feedback. Lee and Moore's (2024) review found that generative-AI feedback systems may support personalized learning, instructional dialogue, and workload reduction, but the design and value of feedback depend on the instructional context, content, mechanism, and role of the instructor.

Reliability and fairness remain critical. Yan et al. (2024) synthesized practical and ethical challenges of large language models in education, including inaccurate outputs, bias, privacy risks, and the need for human oversight. For the present study, AI-assisted assessment will therefore be limited to preliminary analysis or draft feedback based on a predefined rubric. The system may identify whether required elements are present, organize observations by criterion, or suggest areas for revision. It will not issue the final official grade. Facilitators must be able to review, edit, accept, or reject AI suggestions, and the system should preserve an audit trail of final human decisions.

This human-in-the-loop design is especially appropriate for NSTP outputs, which may involve local context, civic values, reflection, feasibility, and community relevance that cannot be reduced to surface text features. A rubric can improve consistency, but professional and educational judgment remains necessary. Assessment quality should thus be evaluated through both technical behavior and user experience: whether the AI follows the rubric, produces relevant and understandable suggestions, avoids unsupported claims, and assists rather than displaces the facilitator.

### *Software Quality and User Acceptance*

A multifunction platform requires a quality model that covers more than correctness. ISO/IEC 25010:2023 defines nine product-quality characteristics applicable to ICT and software products and supports requirements specification, testing, quality control, and acceptance decisions (ISO/IEC, 2023). Functional suitability concerns whether required NSTP functions are complete, correct, and appropriate. Performance efficiency addresses response time, resource use, and capacity. Compatibility considers coexistence and interoperability. Interaction capability addresses how specified users understand, learn, operate, and receive assistance from the interface. Reliability concerns consistent operation and recovery; security concerns protection, accountability, and resistance to unauthorized action; maintainability concerns analysis, modification, and testing; flexibility concerns adaptation, scaling, installation, and replacement; and safety concerns avoiding or controlling harmful consequences.

Not every subcharacteristic will require the same instrument or respondent. IT experts may be best positioned to assess architecture, security controls, maintainability, and compatibility through inspection and testing, while administrators, coordinators, facilitators, and students can evaluate task completion, clarity, ease of learning, usefulness, and satisfaction in realistic workflows. Performance logs, functional tests, security checks, expert ratings, user questionnaires, and interviews can provide complementary evidence. The study should explicitly map each selected characteristic to indicators and evidence rather than treating ISO/IEC 25010 as a single general satisfaction scale.

User acceptance is necessary because a technically sound system can remain unused if users expect little benefit, perceive excessive effort, or lack support. UTAUT2

explains technology acceptance through constructs that include performance expectancy, effort expectancy, social influence, facilitating conditions, hedonic motivation, price value, and habit, with individual differences affecting relationships (Venkatesh et al., 2012). The present study does not adopt the full UTAUT2 model; however, its emphasis on expected usefulness, effort, and support informs the assessment of user perceptions and experiences. For an institutionally provided platform, price value is not central, while workflow usefulness, efficiency, ease, support, and acceptability are directly relevant.

### *Privacy, Security, and Responsible AI*

The Data Privacy Act of 2012 applies to the processing of personal information in information and communications systems and specifically recognizes education-related information as sensitive personal information in relevant circumstances (Republic of the Philippines, 2012). The proposed platform will process identity, enrollment, attendance, grades, submissions, and communication data. Privacy must therefore be designed into data collection, access, storage, use, disclosure, retention, and disposal. Users should receive clear notices, and access should be limited to what each role needs. Sensitive records should be protected in transit and at rest where technically appropriate, and changes to critical records should be attributable to authorized accounts.

AI integration adds risks because prompts or uploaded content may be transmitted to an external model or processed by a separate service. UNESCO (2023) recommends a human-centered approach to generative AI in education that protects human agency, inclusion, privacy, and educational values. Yan et al. (2024) likewise identify practical and ethical concerns involving reliability, explainability, data protection, bias, and misuse. These concerns imply data minimization, clear AI-use boundaries, testing for unsupported

or harmful outputs, disclosure when content is AI generated, and a mechanism to report and correct problematic responses.

Responsible AI in this study means that automation remains subordinate to legitimate educational and administrative purposes. The chatbot should not disclose another user's data; recommendations should not make irreversible decisions; proposal advice should not fabricate evidence; and assessment support should not replace the facilitator. Human review, documentation, access control, auditability, and the ability to override or disable AI functions are therefore part of the platform requirements rather than optional enhancements.

**Synthesis**

The reviewed literature and current institutional context converge on the need for an integrated, quality-assured, and human-centered NSTP platform. CHEDRO III's electronic reporting directives establish the practical importance of structured and timely NSTP data, while TAU's recent CWTS and LTS completion volume demonstrates the scale at which student, attendance, assessment, and completion records must be coordinated (CHEDRO III, 2022a, 2022b, 2024, 2025; Tarlac Agricultural University, 2026). Studies of e-learning system success indicate that accurate information and dependable functions must be accompanied by educational relevance, support, and attention to the needs of learners and instructors (Al-Fraihat et al., 2020). Together, these sources justify a centralized platform that connects administrative and learning processes instead of digitizing them as isolated modules.

The literature also supports the principal technologies proposed in the study, but with qualifications. QR-based attendance can speed up recording and link attendance

events to a server-based record (Rahmawati et al., 2020), yet authentication, time controls, duplicate prevention, facilitator verification, and correction procedures are necessary to protect integrity. Learning analytics can make assessment and activity data useful for dashboards, recommendations, and interventions (Ifenthaler & Yau, 2020; Mangaroska & Giannakos, 2019), but recommendations should be explainable, curriculum aligned, and subject to educator interpretation.

Educational chatbot research provides evidence for immediate assistance, personalized support, administrative information, and advising (Kuhail et al., 2023; Labadze et al., 2023; Okonkwo & Ade-Ibijola, 2021). AI assessment and feedback studies similarly show potential for timely formative support (Gonzalez-Calatayud et al., 2021; Lee & Moore, 2024). However, the same literature identifies reliability, bias, privacy, transparency, and overreliance as continuing concerns (UNESCO, 2023; Yan et al., 2024). The synthesis therefore supports a bounded AI design: a chatbot grounded in approved information, a proposal advisor that scaffolds rather than authors, personalized recommendations tied to observable learning data, and rubric-based feedback that remains subject to facilitator review.

A clear research gap remains in the combination of these functions for the specific operational environment of a Philippine university NSTP office. Existing work commonly studies learning management, attendance, chatbots, analytics, or AI feedback separately. It does not establish whether a unified platform can fit the interdependent registration, component and section management, attendance, learning, grading, communication, and reporting workflows of TAU-NSTP while also meeting current software-quality, privacy, and responsible-AI expectations. The present study addresses this contextual and

integrative gap through requirements assessment, iterative development, technical and user evaluation, and mixed-method analysis.

The synthesis further indicates how the platform should be assessed. ISO/IEC 25010:2023 provides the current structure for product quality, while user-perception literature emphasizes usefulness, effort, and facilitating conditions (ISO/IEC, 2023; Venkatesh et al., 2012). Accordingly, the study will not equate successful development with feature completion alone. It will examine functional and non-functional requirements, technical quality, intended users' perceptions and experiences, and specific safeguards for personal data and AI-assisted decisions. These linked concerns form the basis of the study's conceptual framework and research questions.

**Conceptual Framework**

The study adopts an Input-Process-Output (IPO) framework to organize the development and assessment of the proposed platform. The framework is appropriate for a developmental study because it connects the conditions and requirements entering the project, the systematic activities used to transform them, and the resulting artifact and evidence. It is not treated as a causal theory of learning; rather, it is a project-level conceptual map that ensures alignment among the identified problem, system requirements, development activities, evaluation, and expected output.

The input component includes the existing NSTP workflows for registration and sectioning, attendance, student and facilitator records, grades, learning materials, assessments, communication, and reports. It also includes the needs and constraints of the NSTP administrator, coordinators, facilitators, students, and IT experts; relevant institutional forms and policies; and the technical requirements for a centralized web

platform, QR-based attendance, role-based access, automation, and bounded AI assistance. Privacy, security, and responsible-AI requirements are treated as inputs from the beginning rather than as checks added after development.

The process component begins with documentation and analysis of current practices and stakeholder requirements. The platform is then designed, developed, integrated, tested, reviewed, and refined through iterative cycles. Administrative, learning, reporting, QR, and AI features are validated against requirements and realistic user tasks. The developed system is assessed through functional and technical testing, relevant ISO/IEC 25010:2023 characteristics, user evaluation, and qualitative feedback. Quantitative and qualitative findings are integrated to identify strengths, defects, unmet needs, risks, and recommended revisions.

The output is a developed and assessed Smart NSTP Management and AI-Integrated Learning Platform for the TAU-NSTP Office, together with documented evaluation results and recommendations for improvement and possible implementation. Feedback from testing and assessment is expected to return to the process component for revision; thus, although the IPO diagram is presented linearly for clarity, development is iterative in practice.

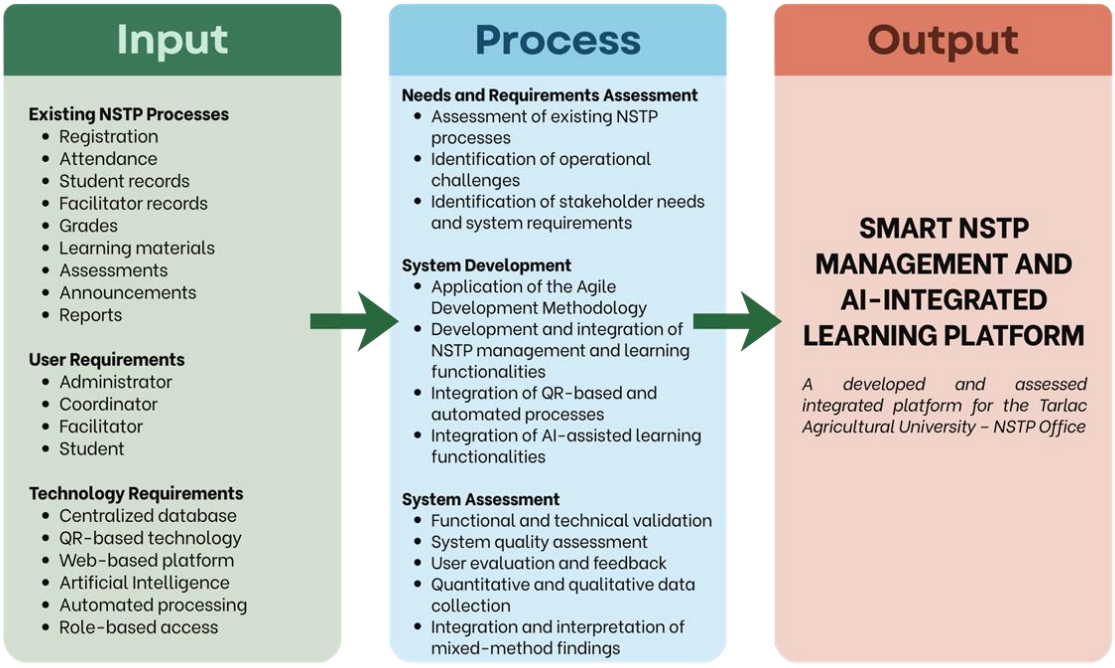



**Figure 1**

*Input-Process-Output Conceptual Framework of the Study*

## Research Problem

The study aims to address the challenges associated with the manual and fragmented management of NSTP administrative and learning processes through the development and assessment of a Smart NSTP Management and AI-Integrated Learning Platform for the Tarlac Agricultural University-NSTP Office. Specifically, it seeks to answer the following questions:

1. What are the existing processes, challenges, and requirements of the TAU-NSTP Office in terms of:

    1.1 student registration and sectioning;

    1.2 attendance monitoring;

    1.3 student and facilitator record management;

1.4 grade computation;

1.5 learning materials and assessments;

1.6 communication and announcements; and

1.7 report preparation?

2. What functional and non-functional requirements should be incorporated into the proposed Smart NSTP Management and AI-Integrated Learning Platform?

3. How can the proposed platform be designed and developed in terms of:

3.1 administrative management;

3.2 student monitoring;

3.3 teaching and learning;

3.4 assessment;

3.5 reporting; and

3.6 AI-assisted learning support?

4. How can AI be incorporated into the platform to provide:

4.1 chatbot-based assistance;

4.2 project proposal guidance;

4.3 personalized learning recommendations; and

4.4 facilitator assessment support?

5. How may the developed platform be evaluated in terms of the relevant ISO/IEC 25010:2023 product-quality characteristics of functional suitability, performance efficiency, compatibility, interaction capability, reliability, security, maintainability, flexibility, and safety?

6. What are the perceptions and experiences of the intended users regarding the usability, usefulness, efficiency, and acceptability of the developed platform?

## Scope and Delimitation

The study covers the analysis, design, development, testing, and assessment of a web-based Smart NSTP Management and AI-Integrated Learning Platform for the Tarlac Agricultural University-NSTP Office. It focuses on the University's CWTS, LTS, and ROTC components and involves NSTP administrators, coordinators, facilitators, enrolled students, and IT experts. The student sample will be determined from the official population at the time of data collection using the procedure specified in Chapter 2.

The platform includes registration and sectioning, student and facilitator records, QR-based attendance, grade management, announcements, and report generation. Its learning functions cover approved materials, quizzes and activities, submissions, scores, feedback, and progress monitoring. Role-based access will limit functions and records according to each user's responsibilities.

AI functions are limited to an NSTP chatbot, project-proposal guidance, personalized learning recommendations, and preliminary assessment feedback subject to facilitator review. They will not approve proposals, assign final grades, impose sanctions, or replace authorized personnel. Assessment covers functional and technical testing, ISO/IEC 25010:2023, and user evaluation. The study excludes university-wide systems, biometrics, financial or personnel administration, and unauthorized external database integration. Findings are limited to the TAU-NSTP setting and tested platform version.

## Significance of the Study

The study may be significant to the following stakeholders:

**TAU NSTP Administrator and Coordinators**. The platform may provide a consolidated view of registration, component and section assignments, attendance, facilitators, grades, announcements, and reports. Validated workflows may reduce repetitive consolidation and improve the timeliness and traceability of required information.

**NSTP Facilitators**. Centralized class lists, attendance tools, materials, assessments, grade computation, communication, and rubric-based AI assistance may reduce clerical work and support more timely feedback. Human-review controls preserve facilitator authority over academic judgments.

**NSTP Students**. Students may gain clearer access to program information, learning materials, announcements, attendance and progress records, feedback, and guided proposal preparation. Transparent AI boundaries and privacy safeguards may also help students use automated assistance responsibly.

**Community Beneficiaries**. Improved administrative coordination and better-guided student project proposals may indirectly support the planning, documentation, and delivery of civic welfare and literacy activities. The study does not, however, measure community impact directly.

**Researchers and System Developers**. The study may contribute a context-specific example of integrating NSTP administration, learning management, QR attendance, reporting, and human-centered AI. Its requirements, evaluation approach, limitations, and findings may guide related educational information-system studies.

## Definition of Terms

The following terms are defined conceptually and, where appropriate, operationally as used in the study:

**AI-Assisted Assessment Support**. A platform function that applies AI to produce preliminary, rubric-based observations or draft feedback on student outputs. The facilitator reviews and retains authority over the official evaluation and grade.

**AI Chatbot**. A conversational interface that responds to NSTP-related questions using approved office information and learning content. In the study, it provides informational assistance and escalation, not authoritative decisions.

**AI-Integrated Learning Platform**. A digital learning environment in which bounded AI functions are embedded into access to information, proposal guidance, resource recommendation, and assessment support.

**Artificial Intelligence**. Computer-based techniques that perform tasks associated with language processing, recommendation, pattern analysis, or decision support. In this study, AI functions assist users and remain subject to defined content, permissions, and human oversight.

**Attendance Management**. The functions for recording, validating, correcting, viewing, and reporting student or facilitator attendance, including time-in and present, late, or absent status.

**Compatibility**. The ISO/IEC 25010:2023 quality characteristic concerning the platform's ability to coexist and exchange usable information with relevant products or environments under specified conditions.

**Facilitator**. An authorized person assigned to deliver NSTP instruction or activities, manage a section, monitor attendance and submissions, assess outputs, and verify grades within the platform

**Flexibility**. The ISO/IEC 25010:2023 quality characteristic concerning adaptability, scalability, installability, and replaceability under specified conditions.

**Functional Suitability**. The extent to which the platform provides complete, correct, and appropriate functions for stated NSTP needs under specified conditions.

**Interaction Capability**. The ISO/IEC 25010:2023 quality characteristic concerning the platform's capacity to support effective user interaction, including learnability, operability, user assistance, inclusion, and protection from user error.

**Learning Analytics**. The collection and interpretation of assessment, activity, and progress data to support feedback, monitoring, and learning recommendations. It is used as decision support rather than an autonomous high-stakes decision mechanism.

**Learning Management**. The platform functions for organizing and distributing learning materials, administering assessments, receiving submissions, recording scores, presenting feedback, and monitoring progress.

**Maintainability**. The ISO/IEC 25010:2023 quality characteristic concerning how effectively and efficiently the platform can be analyzed, modified, tested, and improved.

**National Service Training Program**. The University program implemented through CWTS, LTS, and ROTC components. In the study, NSTP refers specifically to the administrative and learning operations under the TAU-NSTP Office.

**Performance Efficiency**. The ISO/IEC 25010:2023 quality characteristic concerning response time, resource utilization, and capacity relative to specified conditions and workloads.

**Personalized Learning Recommendation**. A system-generated suggestion of facilitator-approved learning content or action based on a student's recorded assessment result, progress, or incomplete requirement.

**Project Proposal Advisor**. An AI-assisted function that scaffolds NSTP proposal development through structured questions and suggestions. It does not invent evidence, determine feasibility, or replace student authorship and facilitator approval.

**QR Code**. A machine-readable two-dimensional code used in the platform to identify an authorized student or attendance transaction and connect the scan to a server-side record.

**Reliability**. The ISO/IEC 25010:2023 quality characteristic concerning consistent operation, availability, fault tolerance, and recovery under specified conditions.

**Role-Based Access**. A security mechanism that grants functions and data access according to the responsibilities of a user role, such as administrator, coordinator, facilitator, student, or technical evaluator.

**Safety**. The ISO/IEC 25010:2023 quality characteristic concerning the platform's ability to avoid or control unacceptable risks and harmful consequences under specified conditions.

**Security**. The ISO/IEC 25010:2023 quality characteristic concerning protection of information and actions through confidentiality, integrity, accountability, authenticity, non-repudiation, and resistance to unauthorized activity.

**Smart NSTP Management Platform**. The integrated, role-based web platform developed in the study to support TAU-NSTP registration, sectioning, attendance, records, learning, assessment, grading, communication, reporting, and bounded AI functions.

**TAU-NSTP Office**. The Tarlac Agricultural University unit responsible for coordinating the University's NSTP components and the principal organizational setting of the study.

**User Acceptability**. The extent to which intended users judge the platform appropriate and are willing to use it for relevant NSTP tasks, considering usefulness, effort, efficiency, support, trust, and fit with their responsibilities.

# Chapter 2

# RESEARCH METHODOLOGY

This chapter presents the methodology for developing and assessing the Smart NSTP Management and AI-Integrated Learning Platform for the Tarlac Agricultural University-National Service Training Program (TAU-NSTP) Office. It describes the research design, system-development procedure, locale, respondents and sampling plan, instruments, data-gathering procedure, data-analysis techniques, and ethical safeguards. The methodology is structured to produce a usable software artifact while generating evidence about its functional quality, user experience, acceptability, and bounded use of artificial intelligence.

## Research Design

The study will use a developmental design-science approach with an embedded mixed-method evaluation. Design science is appropriate because the investigation begins with an identified organizational problem, specifies stakeholder and technical requirements, develops an artifact, demonstrates it in its intended context, and evaluates whether it addresses the problem. These activities correspond to the design-science framework of problem explication, requirements definition, artifact design and development, demonstration, and evaluation (Johannesson & Perjons, 2014). The primary artifact will be the working web-based platform, supported by its requirements specification, design documentation, test evidence, evaluation results, and recommendations for improvement.

Development and evaluation will be iterative rather than strictly linear. Stakeholder findings will inform the initial requirements; prototypes and working increments will be

reviewed; test results and user feedback will be converted into revisions; and the revised version will be retested before final assessment. The life-cycle activities will be aligned with the common framework of ISO/IEC/IEEE 12207:2026, which allows software processes to be applied iteratively and incrementally across development, operation, support, and improvement (International Organization for Standardization et al., 2026). The standard will guide process organization, but the study will not claim formal organizational certification.

The evaluation component will combine quantitative and qualitative evidence. Quantitative data will include test-case results, defect counts, task outcomes, and structured ratings of product quality, usability, usefulness, efficiency, and acceptability. Qualitative data will be obtained from interviews, observation notes, open-ended questionnaire responses, and evaluator comments. A mixed-method approach permits numerical patterns to be explained by the experiences and context reported by users (Creswell & Plano Clark, 2018). The two evidence strands will be interpreted together during evaluation and revision.

The study is non-experimental because participants will not be randomly assigned to competing interventions and the research questions do not seek to establish a causal effect. No statistical hypothesis will therefore be tested in the planned evaluation. The unit of analysis will be the developed platform as used by authorized stakeholder groups for representative TAU-NSTP tasks. Evaluation results will describe the tested version and context rather than claim long-term learning gains, organization-wide adoption, or superiority over every alternative system.

## System Development Procedure

The study will apply the Agile Software Development Methodology through six iterative phases: Analysis, Design, Develop, Testing, Implement, and Review and Refinement. Each phase will produce reviewable outputs and may return to an earlier phase when requirements, defects, or risks require correction. A version-controlled repository and change log will preserve traceability between stakeholder needs, implemented features, test cases, defects, and revisions. Figure 2 presents the development cycle used in the study.

**Analysis.** The researcher will document current NSTP workflows, records, reports, learning activities, pain points, and control requirements through interviews, observation, and review of authorized forms or reports. Stakeholder needs will be translated into functional and non-functional requirements, user roles, permissions, business rules, report requirements, acceptance criteria, and an initial product backlog. Privacy, security, accessibility, data quality, backup, audit logging, and human-review requirements will be defined before implementation.

**Design.** The researcher will prepare the database schema, process models, role-permission matrix, system architecture, navigation structure, wireframes, report layouts, and testable interface designs. The design will separate presentation, application, data, and AI-service concerns where practical to improve maintainability and control access. Representative users will review prototypes for clarity, task fit, terminology, and missing functions before full development.

**Develop.** Core administrative features will be implemented first, followed by learning, assessment, reporting, communication, QR, and AI-assisted functions. Each working increment will undergo developer checks before integration. AI functions will connect only to approved information sources and defined tasks. Prompts, outputs,

fallback messages, and escalation paths will keep suggestions reviewable and prevent them from becoming autonomous official decisions.

**Testing.** Unit, integration, system, permission, data-validation, report, backup-recovery, and acceptance tests will be executed against documented requirements. Critical workflows will include registration, section assignment, attendance capture and correction, score and grade handling, content access, report generation, and account authorization. Defects will be classified by severity, corrected, and retested before implementation proceeds.

**Implement.** A stable build will be deployed in a controlled TAU-NSTP setting using representative accounts and approved data. Eligible participants will complete role-appropriate tasks while the researcher records task outcomes, technical observations, and operational issues. No controlled pilot will proceed while a known critical defect could expose records, corrupt official data, bypass authorization, or produce an irreversible academic decision.

**Review and Refinement.** Technical findings, user ratings, qualitative feedback, and operational logs will be reviewed together. Agreed corrections will be added to the backlog, implemented in the next iteration, and regression-tested. The cycle will continue until the evaluated release satisfies the approved requirements and the researcher can prepare final implementation recommendations.

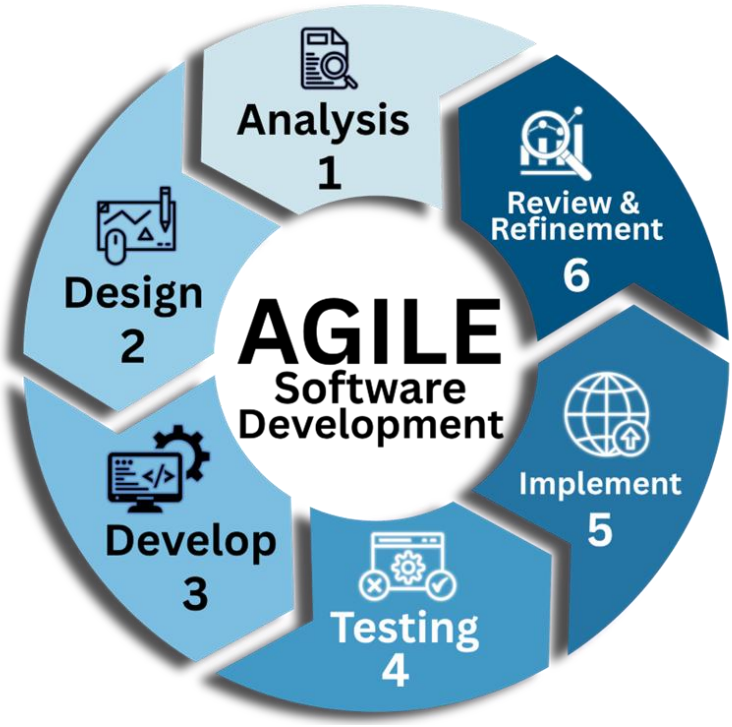



**Figure 2**
*Agile Software Development Cycle of the Study*

**AI Risk Controls and Evaluation**

The chatbot, project-proposal advisor, personalized recommendations, and preliminary assessment feedback will be treated as decision-support features. Their evaluation will consider groundedness in approved sources, relevance, clarity, consistency, refusal of prohibited requests, privacy, traceability, and successful escalation to a facilitator or authorized office personnel. Test prompts will include normal requests, ambiguous questions, requests containing unnecessary personal data, attempts to obtain prohibited decisions, and questions outside the approved knowledge scope. The test set and expected behavior will be documented before scoring.

Responsible-AI controls will be informed by UNESCO's human-centered guidance for generative AI in education and the NIST Generative AI Profile, which emphasizes governance, mapping, measurement, and management of risks across the AI life cycle

(National Institute of Standards and Technology, 2024; UNESCO, 2023). AI-generated content will be visibly identified as assistance, material outputs will require human review, and evaluators will record unsupported, unsafe, biased, or privacy-inappropriate responses. The study will not use AI output as the sole basis for a final grade, proposal approval, disciplinary action, or permanent student classification.

**Locale of the Study**

The study will be conducted at Tarlac Agricultural University in Camiling, Tarlac, with the TAU-NSTP Office as the principal organizational setting. The office coordinates the University's Civic Welfare Training Service (CWTS), Literacy Training Service (LTS), and Reserve Officers' Training Corps (ROTC) components. University reports of orientations, training activities, and more than one thousand recent CWTS and LTS completers demonstrate that the setting includes multiple components, sections, facilitators, learning activities, and substantial student records (Tarlac Agricultural University, 2026).

Requirements gathering will occur in the actual office and instructional context when authorized. Development may be performed on a secured local workstation or approved development environment. Testing will use a separate test instance before any controlled pilot, and real records will not be copied into development tools unless formally authorized and adequately protected. The locale is appropriate because the proposed platform is tailored to TAU-NSTP workflows, terminology, roles, reports, and institutional controls.

## Respondents

The study will involve stakeholder groups that can provide operational, user, and technical evidence. These groups are the NSTP administrator, NSTP coordinators, NSTP facilitators, currently enrolled NSTP students, and information-technology experts. A participant will be invited only for activities relevant to the person's role; for example, students will not evaluate administrative functions they are not authorized to access, while technical experts will focus on product-quality and control characteristics.

**Table 1**

*Participant Groups and Sampling Plan*

| Participant Group | Selection Approach | Primary Contribution | Eligibility |
|---|---|---|---|
| NSTP administrator and coordinators | Complete enumeration or purposive inclusion | Validate workflows, rules, reports, roles, and administrative acceptance | Currently assigned to TAU-NSTP operations and willing to participate |
| NSTP facilitators | Complete enumeration when feasible; otherwise purposive by component | Evaluate section, attendance, learning, assessment, grading, and communication tasks | Currently handling a CWTS, LTS, or ROTC section |
| NSTP students | Proportionate stratified random sampling by component and, when feasible, section | Perform student tasks and evaluate usability, usefulness, and acceptability | Currently enrolled in NSTP, eligible for consent, and has used the pilot version |
| IT experts | Purposive sampling; at least five qualified evaluators when available | Assess technical behavior and relevant ISO/IEC 25010:2023 characteristics | Relevant degree or professional experience in systems development, software quality, security, databases, or educational technology |

Personnel groups with small populations will be invited through complete enumeration when practical because each role may hold distinct process knowledge. Purposive selection may be used when only specific members have direct responsibility for a component, report, or technical area. Information-technology experts will be selected on documented qualifications rather than convenience alone. This combination is

appropriate for a context-specific developmental study in which operational knowledge and technical competence are essential.

For the student survey, the official eligible enrollment at the time of data collection will be used as the population (N). The initial sample will be estimated for a proportion at 95% confidence, 5% margin of error, and maximum variability (p = .50), then adjusted for the finite population. These values may be revised with justification if the ethics-approved protocol or accessible population requires a different precision level. Adequate and transparent sample determination helps reduce avoidable sampling error and bias (Taherdoost, 2017).

*Initial sample: n0 = [Z^2 x p(1 - p)] / e^2*

*Finite-population adjustment: n = n0 / [1 + (n0 - 1) / N]*

*Proportionate allocation: nh = (Nh / N) x n*

In the formulas, Z is the standard value for the selected confidence level, p is the estimated population proportion, e is the margin of error, N is the eligible student population, Nh is the population of a component or stratum, and nh is its allocated sample. A reserve list may be drawn within each stratum to address nonresponse without replacing participants from a different component. Students who have not used the evaluated build, are no longer enrolled, decline consent, or submit substantially incomplete responses will be excluded from the corresponding analysis. Reasons for exclusion and final response counts will be reported.

**Research Instrument**

The study will use complementary instruments because no single measure can establish that a multi-role information system is operationally appropriate, technically

sound, usable, and responsibly AI-enabled. Each instrument will be linked to specific research questions, participants, evidence, and analysis procedures. Instrument versions, validation decisions, and administration instructions will be documented to support consistency and auditability.

**Table 2**

*Research Instruments and Evidence Produced*

| Instrument | Source/Respondents | Evidence Produced |
|---|---|---|
| Requirements interview guide and workflow checklist | Administrator, coordinators, facilitators; authorized records and forms | Current processes, pain points, roles, rules, data fields, reports, risks, and prioritized requirements |
| Functional and acceptance test cases | Researcher, selected process owners, and authorized testers | Expected versus actual results, pass/fail status, defects, severity, corrections, and retest results |
| Product-quality questionnaire | IT experts and relevant process owners | Ratings and comments for applicable ISO/IEC 25010:2023 quality characteristics |
| User-experience and acceptability questionnaire | Administrators, coordinators, facilitators, and students after task use | Ratings of effectiveness, efficiency, satisfaction, usefulness, effort, support, trust, and intention to use |
| AI behavior evaluation checklist | Facilitators, domain personnel, and IT experts | Groundedness, relevance, boundary compliance, privacy, safety, escalation, and error observations |
| Open-ended evaluation guide | All consenting evaluator groups | Strengths, difficulties, unmet needs, risks, and recommended changes |
| System and test logs | Generated by the controlled test environment | Task completion, errors, response events, access events, and evidence for defect analysis without unnecessary personal data |

**Requirements Instruments.** The semi-structured interview guide will cover existing workflows, records, reports, user roles, pain points, exceptions, and desired outcomes. The workflow checklist will help verify the sequence, responsible role, input, validation, output, and control point for each process. With permission, the researcher may inspect blank or redacted versions of forms and reports; collection of unrelated personal information will be avoided.

**Functional and Technical Instruments.** Test cases will state the requirement identifier, preconditions, input, procedure, expected result, actual result, pass/fail status, evidence, defect severity, correction, and retest result. The product-quality questionnaire will operationalize the applicable ISO/IEC 25010:2023 characteristics of functional suitability, performance efficiency, compatibility, interaction capability, reliability, security, maintainability, flexibility, and safety (International Organization for Standardization & International Electrotechnical Commission, 2023). Not every item will be rated by every user group; technical or internal characteristics will be assigned to qualified evaluators.

**User Evaluation Instrument.** After completing assigned tasks, intended users will rate effectiveness, efficiency, satisfaction, clarity, learnability, usefulness, facilitating conditions, trust, and willingness to use the platform. Usability items will reflect effectiveness, efficiency, and satisfaction in a specified context of use under ISO 9241-11:2018 (International Organization for Standardization, 2018). Acceptability items will be adapted to the institutional context from relevant technology-acceptance constructs, particularly performance expectancy, effort expectancy, facilitating conditions, and behavioral intention (Venkatesh et al., 2012).

**AI Evaluation Instrument.** The AI checklist will use predetermined scenarios and expected boundaries. Evaluators will record whether a response is supported by approved content, relevant to the request, understandable, appropriately qualified, free of unnecessary personal-data exposure, and correctly escalated when the request is outside scope or requires human authority. Each serious unsupported or harmful response will be documented as an issue even when an average rating appears favorable.

## Validation and Reliability of Instruments

The researcher-developed interview guides, checklists, questionnaires, and task scenarios will be submitted to a panel of subject-matter, research-method, and information-technology experts. They will examine relevance, clarity, representativeness, alignment with the research questions, and appropriateness for each respondent group. Item-level and scale-level content-validity indices may be computed from expert ratings, and unclear or weakly relevant items will be revised or removed. The procedure will follow transparent content-validation practices described by Yusoff (2019).

The revised questionnaires will be pilot-tested with individuals similar to the intended respondents but excluded from the main evaluation whenever feasible. Pilot feedback will be used to refine instructions, item wording, response options, task duration, and administration sequence. Cronbach's alpha will be computed separately for coherent multi-item scales, interpreted together with item behavior and the construct definition rather than treated as proof of validity. A value near or above .70 will be considered generally acceptable for an exploratory instrument, but the number of items, dimensionality, and context will also be reported because alpha can be misinterpreted when used mechanically (Taber, 2018).

## Data Gathering Procedure

**Approval and Coordination.** Before recruitment or access to records, the researcher will obtain approval from the adviser, research or ethics authority, TAU administration, and the NSTP Office as applicable. The approved protocol will define the participant groups, instruments, data requested, system environment, privacy

responsibilities, and authorized personnel. No operational deployment will be represented as institutional approval beyond the permissions actually granted.

**Requirements Gathering.** Consenting key informants will participate in interviews or focused process-review sessions. Authorized workflows and documents will be examined, and findings will be summarized in process descriptions and a requirements traceability matrix. Participants will be asked to verify the accuracy of the documented current process and the priority of proposed requirements.

**Design and Development.** Validated requirements will be converted into prototypes, architecture, database design, user stories or work items, acceptance criteria, and test cases. The platform will be developed in increments, reviewed by relevant stakeholders, and revised. Test data will be synthetic, de-identified, or specifically authorized. Development logs will document decisions and changes.

**Instrument Validation and Pilot Testing.** Experts will review the research instruments before administration. The revised instruments and task scripts will then be pilot-tested. Problems in wording, scale consistency, navigation, task order, time demand, or data capture will be corrected before the main evaluation. Pilot responses will not be combined with the final dataset unless the protocol explicitly permits this and no material change was made.

**Technical and Functional Evaluation.** The researcher and authorized evaluators will execute the test suite in a controlled environment. Test evidence will include the build identifier, environment, tester, date, expected and actual result, supporting screenshot or log when appropriate, defect severity, and retest status. Role permissions and critical record operations will be tested with both authorized and unauthorized scenarios. AI scenarios

will be run using the same documented version and settings whenever practical to support reproducibility.

**User Evaluation.** Eligible participants will receive a brief orientation, consent information, and role-specific task script. They will use the platform individually or in supervised sessions without coaching that would conceal usability problems. After task completion, participants will answer the appropriate questionnaire and may provide open-ended comments or join a short interview. Participation or ratings will not affect grades, enrollment, employment, or access to NSTP services.

**Consolidation and Revision.** Quantitative results, qualitative feedback, logs, and defects will be consolidated by research question and feature. The researcher will distinguish required corrections from desirable future enhancements, revise the system within the approved scope, and perform regression testing. The final report will identify the tested build, unresolved limitations, and recommendations for any subsequent institutional implementation.

## Data Analysis Technique

Quantitative and qualitative data will be analyzed separately first and then integrated. The analysis will retain respondent-group distinctions because administrators, facilitators, students, and IT experts evaluate different tasks and characteristics. Missing responses will not be silently imputed; the number of valid responses for each analysis will be reported. Items marked not applicable will be excluded from the corresponding mean rather than scored as a neutral response.

**Profile and Response Data.** Frequency and percentage will summarize respondent group, NSTP component, relevant experience, and completion status. Only demographic

or role variables necessary for interpreting the research questions will be collected and reported. Small cells that could reveal an individual's identity will be combined or suppressed.

**Functional and Technical Results.** Test cases will be summarized by module, requirement, result, and defect severity. Pass rate will be computed as the number of passed test cases divided by the number executed, multiplied by 100. A percentage will not override the criticality of a failure: any unresolved defect affecting authorization, privacy, official records, irreversible decisions, or core availability will be discussed individually. AI results will additionally report boundary failures and serious unsupported or unsafe outputs by scenario type.

*Test pass rate (%) = (number of passed test cases / number of executed test cases) x 100*

**Questionnaire Data.** For each coherent item set, the mean and standard deviation will summarize central tendency and response dispersion. Item-level distributions will also be retained so that a favorable mean does not hide a subgroup concern or polarized responses. Reliability will be estimated with Cronbach's alpha for applicable multi-item scales. Results will be presented by respondent group and by quality characteristic where the participants are qualified to judge it.

**Table 3**

*Interpretation of Five-Point Evaluation Ratings*

| Scale | Mean Range | Verbal Interpretation |
|---|---|---|
| 5 | 4.21-5.00 | Strongly Agree / Very High |
| 4 | 3.41-4.20 | Agree / High |
| 3 | 2.61-3.40 | Neither Agree nor Disagree / Moderate |
| 2 | 1.81-2.60 | Disagree / Low |
| 1 | 1.00-1.80 | Strongly Disagree / Very Low |

The verbal ranges will support concise reporting but will not substitute for examination of item wording, standard deviations, comments, task evidence, and defect severity. The study will avoid claiming that equal numerical intervals prove equal psychological distance between response categories; the ratings will be treated as practical summaries for the specified evaluation.

**Qualitative Data.** Interview transcripts or detailed notes and open-ended responses will be analyzed through reflexive thematic analysis. The researcher will become familiar with the data, generate codes, develop and review candidate themes, refine theme definitions, and select evidence that explains the findings. An audit trail will record coding and theme decisions. The analysis will distinguish recurring concerns from isolated but high-risk incidents and will follow quality principles for reflexive thematic analysis (Braun & Clarke, 2021).

**Mixed-Method Integration.** A joint evidence matrix will align research questions, quantitative results, qualitative themes, test evidence, and resulting system decisions. Convergence will be noted when evidence sources support the same conclusion; complementarity when qualitative data adds explanation; and divergence when sources conflict. Divergent results will be investigated rather than averaged away. The integrated findings will determine required corrections, accepted limitations, and recommendations for future implementation.

## Ethical Concerns

The study will begin only after the required institutional and ethics approvals have been obtained. Each participant will receive clear information about the study's purpose, activities, time commitment, foreseeable risks and benefits, voluntary nature,

confidentiality measures, and contact persons. Consent will be documented before data collection. Participants may decline a question or withdraw without penalty, and refusal will not affect academic standing, grades, employment, or access to University services.

Recruitment will minimize power pressure. When students or employees are invited, consent and evaluation data will be collected in a manner that avoids disclosure of individual participation or ratings to a person who directly grades or supervises them whenever feasible. Reports will use aggregated results and non-identifying quotations. Any code linking identities to responses will be stored separately from the analysis dataset and accessed only by authorized research personnel.

Personal information will be processed according to the principles and lawful requirements of the Data Privacy Act of 2012 (Republic of the Philippines, 2012) and applicable University policies. Only data necessary for the research and system test will be collected. Data will be protected in transit and at rest using controls appropriate to the approved environment, including role-based access, strong authentication, restricted exports, secure backup, and audit logging for critical actions. The final report will not expose student numbers, credentials, private messages, raw prompts containing personal data, or identifiable academic records.

Research files will be retained only for the period approved by the ethics authority and required by institutional policy, after which identifiable copies will be securely destroyed or irreversibly de-identified. Development, test, and production-like environments will be separated. Synthetic or de-identified data will be preferred, and any authorized restoration of data for testing will be logged and limited to the minimum needed. Security or privacy incidents will be reported through the approved institutional procedure.

AI-assisted features will be disclosed to users and evaluated with human oversight. Unnecessary personal or sensitive information will not be sent to an external AI service. The researcher will document the provider or model version when available, information sources, prompt boundaries, known limitations, and human-review requirements. AI outputs used as research data will be treated as potentially inaccurate and will be verified against approved sources. Participants will not be asked to surrender authorship, and AI will not make final academic or disciplinary decisions.

The researcher will maintain methodological transparency by reporting the final sample, exclusions, instrument revisions, tested software build, unresolved defects, deviations from the approved procedure, and limitations. Conflicts of interest and the researcher's dual role as developer and evaluator will be acknowledged. Where feasible, independent experts and stakeholder validation will be used to reduce self-evaluation bias.

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